

Riccardo Barbieri

AI Engineer



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Technical Skills

ML NLP MLOps Deep Learning

Data Pipelines LLM Workflows RAG

Experimentation Evaluation

Programming: Python, SQL, MATLAB, R

ML & DL: Scikit-learn, PyTorch, spaCy, Transformers

LLM & RAG: LangChain, LangGraph, Pocketflow, Vector DBs

Data Pipelines: Pandas, NumPy, Spark, Databricks

Cloud & DevOps: AWS, Azure, Docker, Git

Transferable Skills

Project Management:

I manage complex projects involving ML, statistical modeling and AI. I collaborate with multidisciplinary teams, adapting to changing contexts to achieve ambitious goals, even with limited resources.

Creative Thinking and Problem Solving:

I develop innovative solutions for LLM-based systems such as RAG and agents, ML applications, predictive models and data pipelines. I combine an analytical approach with an experimental mindset, with a strong focus on scalability and generalizability.

Interpersonal Skills:

I collaborate effectively with research, data & analytics, data science, and AI engineering teams. I support junior colleagues and facilitate knowledge transfer. I am regarded as a positive and optimistic team member who helps build constructive working relationships.

Work Experience

BitBang (2021 - Ongoing)

Bologna

AI Engineer (2024 - Ongoing)

Development, design, and deployment of systems based on LLM and ML.

- **Enterprise LLM workflows:** Technical lead in the development of LLM-based systems (RAG and agents). Designed and implemented ingestion and retrieval pipelines for multiple products across different business units. Coordinated a cross-functional team to support the progressive evolution of the system into an enterprise AI platform.
- **Automation & classification:** Responsible for the development of a purchase-request classification system. Conducted the system analysis, design, and initial implementation, followed by delegation of activities and mentoring of a junior colleague.
- **Cash flow forecast:** Re-engineered preprocessing, training, forecasting, and output pipelines to enhance scalability, reproducibility, and system reliability.

Data Scientist (2021 - 2024)

Implementation of data-driven solutions based on statistical and ML models.

- **Customer analytics & demand forecasting:** Contributed to the design of a forecasting system for product mix planning. Performed advanced analyses of user navigation data from e-commerce platforms to support marketing and personalization strategies.
- **Marketing performance measurement:** Developed tools for Marketing Mix Modeling based on open-source libraries. Performed attribution analyses including online and offline channels.
- **NLP & Deep Learning:** Developed a custom aspect-extraction model in PyTorch and deployed it in a prototype AWS web app for aspect-based sentiment analysis. Experience applying open-source NLP models.

Charité / BCCN (2016 - 2021)

Berlin

Researcher in Cognitive Neuroscience (2017 - 2021)

Experimental research on brain signals related to decision-making.

- **Experimental design & data analyses:** Designed and conducted behavioral and neuroimaging experiments on human participants. Developed data analysis pipelines using statistical and ML models.
- **Results dissemination:** Authored peer-reviewed publications, presented research at international conferences. Published anonymized datasets and reproducible code for the research community.

Research Intern (Feb 2016 - Sep 2016)

Computational neuroscience research conducted for master's thesis abroad.

- **Neuroimaging data analysis:** Experimentation with ML techniques and development of a classification pipeline starting from an existing dataset.
- **Lab work & programming:** MATLAB programming and collaboration within an interdisciplinary research team.

Education

Doctoral Degree

Charité - Universitätsmedizin Berlin

Master in Cognitive Neuroscience

Università di Padova

Bachelor in Psychology

Università di Bologna